Territory Briefing: Enna

Professional province-level screening based on ARCUS historical evidence, territorial exposure, ARCUS/AINOP Collapse Rate and historical-pattern reading.

PRIORITY
62/100

COLLAPSE
133.44x / national rank 2

EVENTS
3

DECISION MESSAGE
Distributed chronic risk

REPORT DATE
Jun 10, 2026

CONTEXT
Bridge

PROVINCE
Enna

OUTPUT
New construction

ARCUS is built on the first systematic database of collapsed bridges in Italy from 2000 to today: the report does not merely rank cases, it reads documented historical patterns to guide preliminary checks.

01 EXECUTIVE SUMMARY

The selected province requires preliminary screening for Bridge interventions. The value is not the case list: it is understanding whether Enna has already shown concentrated vulnerability under extreme conditions or risk distributed over time.

RECOMMENDED USE

ARCUS reading: for new Bridge interventions, the Hydraulic signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Enna.

PRIORITY INDEX

62/100

exposure index + heritage reading

COLLAPSE RATE

133.44x

ARCUS/AINOP national benchmark; confidence very low

HISTORICAL EVENTS

3

Hydraulic dominant

RELIABILITY

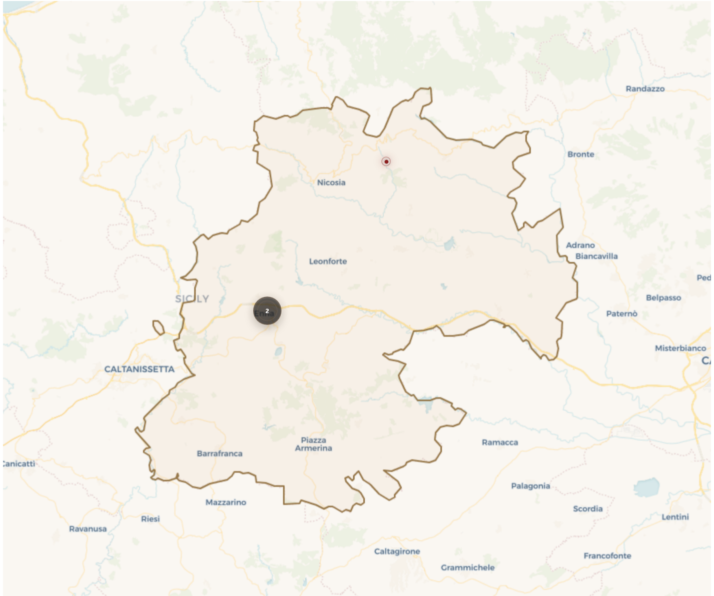
79/100

7 linked sources

COLLAPSE RATE INTERPRETATION

The 133.44x rate versus the national average places Enna among the provinces with documented historical vulnerability above the ARCUS/AINOP benchmark. Enna is national rank 2 by documented ARCUS/AINOP collapse rate. Collapse Rate confidence: very low. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

02 TERRITORY READING & MAP



ENNA / ARCUS ATLAS EXTRACT

GEOMORPHOLOGY AND CONTEXT

Enna shows a hydraulic profile that should be read by geomorphological setting, not as a uniform signal: 3 ARCUS events across 2 municipalities indicate that confined or torrential watercourses, valley crossings and lowland river reaches must be separated during verification.

MAP READING

The map locates 3 ARCUS cases across 2 municipalities and should be read with the historical pattern: Distributed chronic risk. Dominant date: 2015-03-28.

03 HISTORICAL FAILURE CONTEXT

HISTORICAL PATTERN

Distributed chronic risk. historical risk distributed across multiple events and dates (33% on the most recurrent date).

DOMINANT TRIGGER

67%

Hydraulic

HAZARD

hydraulic

dominant territorial signal

READING

distributed risk

33% on the most recurrent date; read as distributed temporal risk plus local event clusters.

04 ACTIONS

- Use the territorial exposure indicators to identify the dominant Hydraulic signal.
- Read the documented distributed historical pattern and use individual cases as map/appendix references.
- For hydraulic-triggered collapses in Enna, priority data includes: hydraulic model with TR100/TR200 flood scenarios, preliminary scour depth estimate at piers, riverbed/bathymetry evidence at crossings and debris-transport indicators where mountain, confined-channel or minor-torrent basins are involved.
- Differentiate checks by geomorphology in Enna: torrential, confined or steep watercourses require scour susceptibility and debris-transport checks; lowland river crossings require flood level verification, inundation duration and backwater sensitivity.
- Use CSV and GeoJSON exports for technical coordination.

DATA REQUEST PACKAGE

TR100/TR200 hydraulic model; preliminary scour depth; riverbed/bathymetry evidence; debris transport indicators where relevant.

COLLAPSE RATE CONFIDENCE NOTE

Collapse Rate confidence: very low. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

05 EVENT REFERENCE APPENDIX

The following list is a documentary reference, not an operational P1/P2/P3 ranking. The decision message remains the provincial historical pattern.

ID	Date	Municipality	Severity	Cause
B15.03.03	Mar 28, 2015	Enna	Total Collapse	Hydraulic
B09.02.01	Feb 01, 2009	Enna	Total Collapse	Hydraulic
B15.03.02	Mar 14, 2015	Cerami	Partial Collapse	Material

06 METHODOLOGY SNAPSHOT

Layer	Meaning
Priority Index	Territorial exposure index: synthesizes dominant hazard, ARCUS case density, event concentration and evidence reliability. It does not directly measure historical bridge-stock vulnerability.
Collapse Rate	Ratio between ARCUS cases and AINOP counted bridges in the province. Value: 133.44x, national rank 2; it should be read separately from the Priority Index.
Why Separate	Priority Index indicates where the territory has elevated hazard exposure; Collapse Rate indicates where the bridge stock has shown historical vulnerability above or below the national average.
Historical Failure Context	Historical-pattern reading: concentrated event or distributed risk over time.
Hydraulic Cause Category	Hydraulic cause category: ARCUS uses 'hydraulic' as the documented trigger when sources confirm a flood or high-water event as the proximate cause. Mechanism-level classification - scour, bank erosion, debris impact, overtopping - is applied only where primary technical sources provide
Evidence Window	Observed period: ARCUS database of collapsed bridges in Italy from 2000 to today; close events in the same time window are read as one scenario, not decision duplicates.
AINOP Coverage	Collapse Rate uses AINOP counted bridges as provincial denominator. Confidence: very low; local undercounting may inflate the ratio.
Evidence Classes	A = primary institutional/technical source; B = reliable technical or media source; C = generic documentary evidence; D = weak evidence to strengthen.